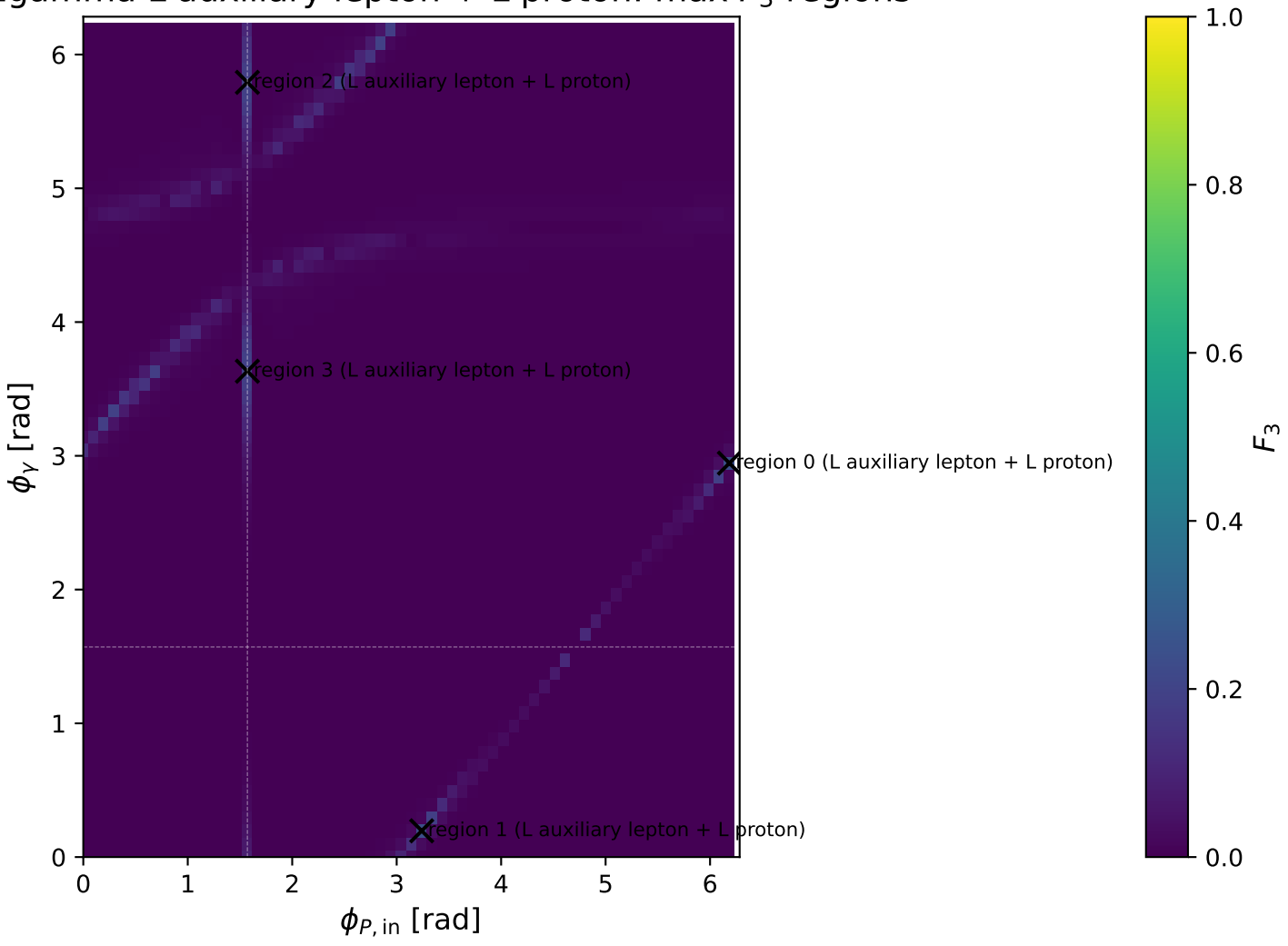
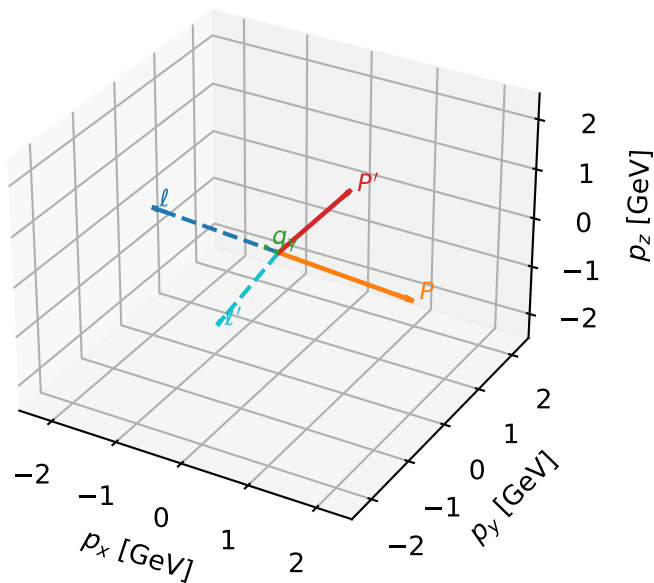


gamma L auxiliary lepton + L proton: max F_3 regions



L auxiliary lepton + L proton characteristic max F_3 configuration

3D momenta

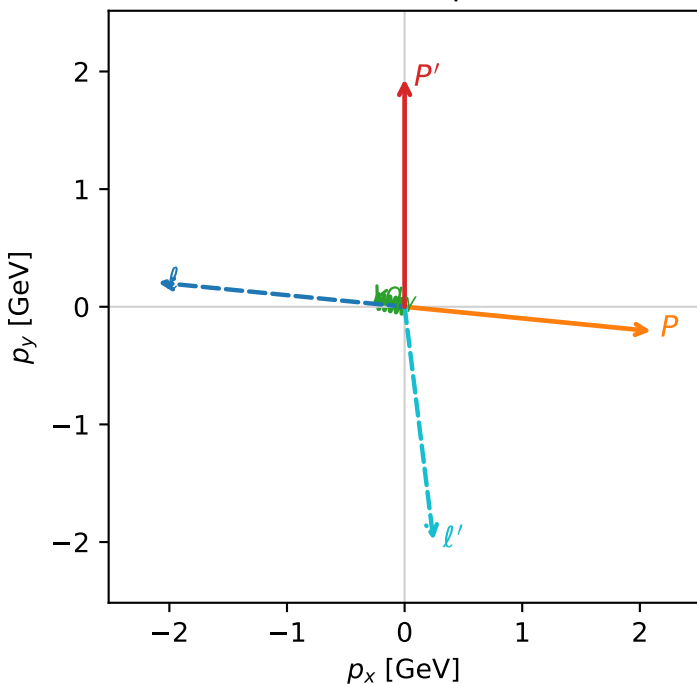


f3_low_egamma_ll_region_0 F_3=0.239851
 region: low_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_l\rangle \otimes |L_p\rangle$

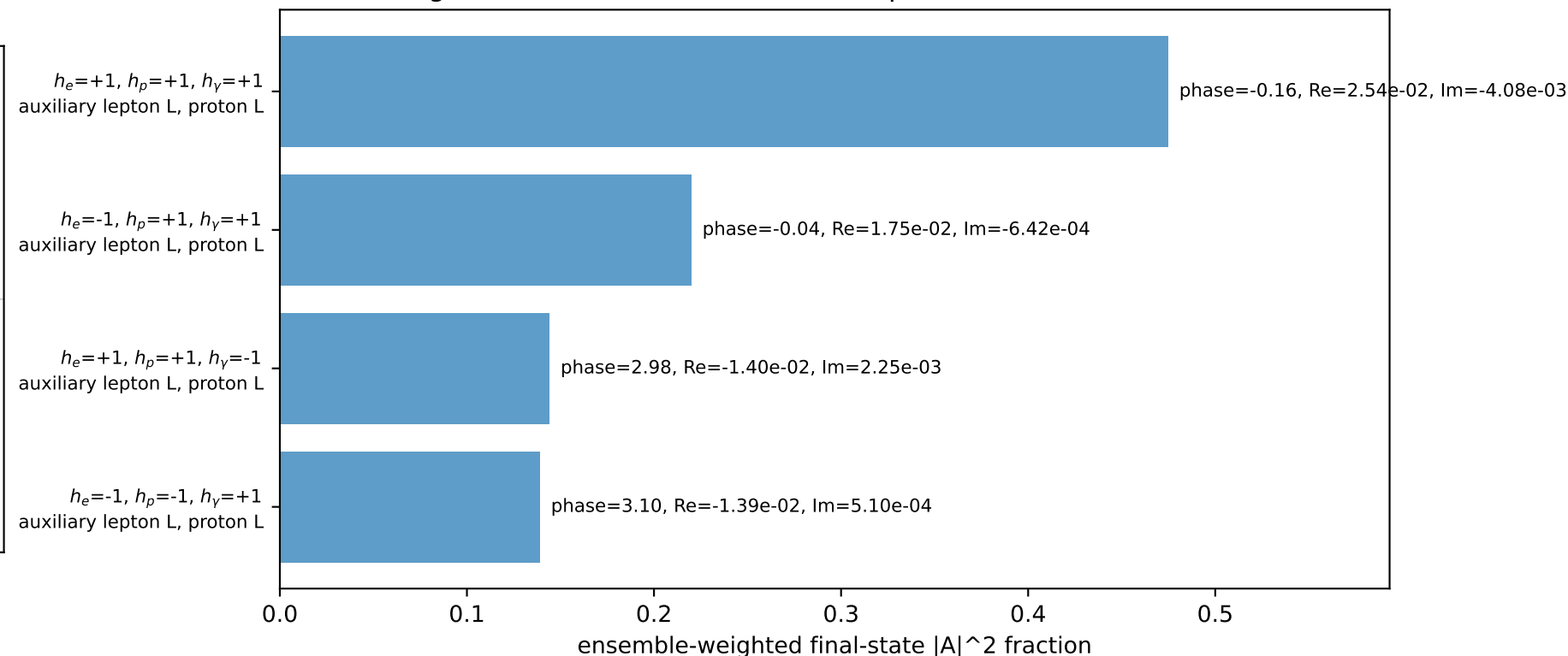
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.93673$
 $\theta_{in}=1.57079$, $\phi_l=3.04342$, $\phi_P=6.18501$
 $E_Y=0.25$, $\phi_Y=2.94524$
 $Q^2=10.2807$, $x_B=0.920571$, $t=-8.96624$, $W^2=1.76689$, $y=0.568741$
 $\theta(l', \gamma)=108.29$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -2.0968$ $p_y= 0.2065$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 2.0968$ $p_y= -0.2065$ $p_z= 0.0000$
 l' $E= 2.2366$ $p_x= 0.2452$ $p_y= -1.9855$ $p_z= 0.0000$
 P' $E= 2.1519$ $p_x= 0.0000$ $p_y= 1.9367$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.2452$ $p_y= 0.0488$ $p_z= 0.0000$

Transverse plane

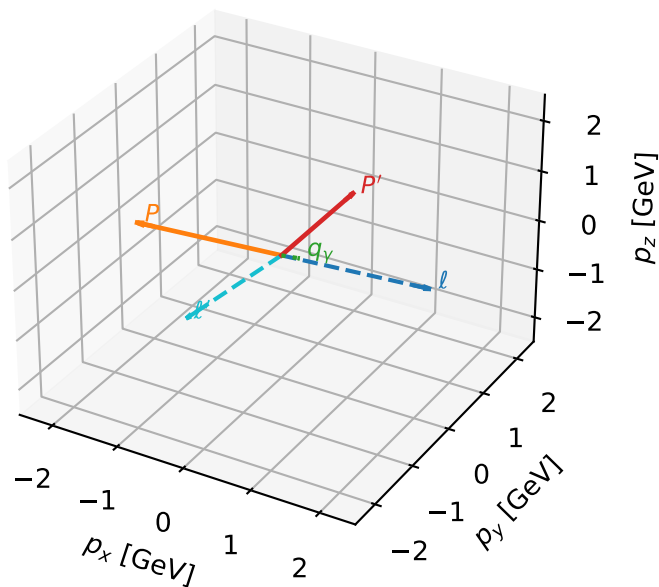


Leading initial-ensemble/final-state components (N=4, retained=97.7%)



L auxiliary lepton + L proton characteristic max F_3 configuration

3D momenta

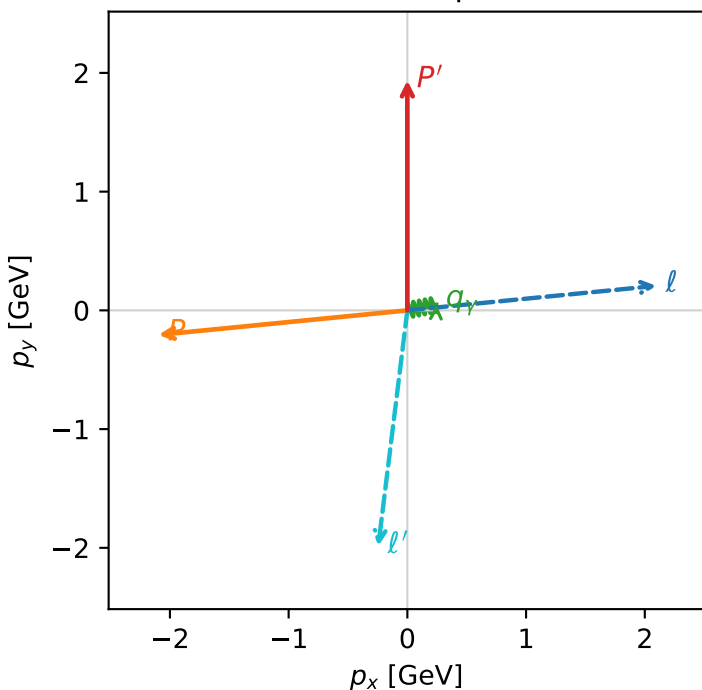


f3_low_egamma_ll_region_1 $F_3=0.239851$
 region: low_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_l\rangle \otimes |L_p\rangle$

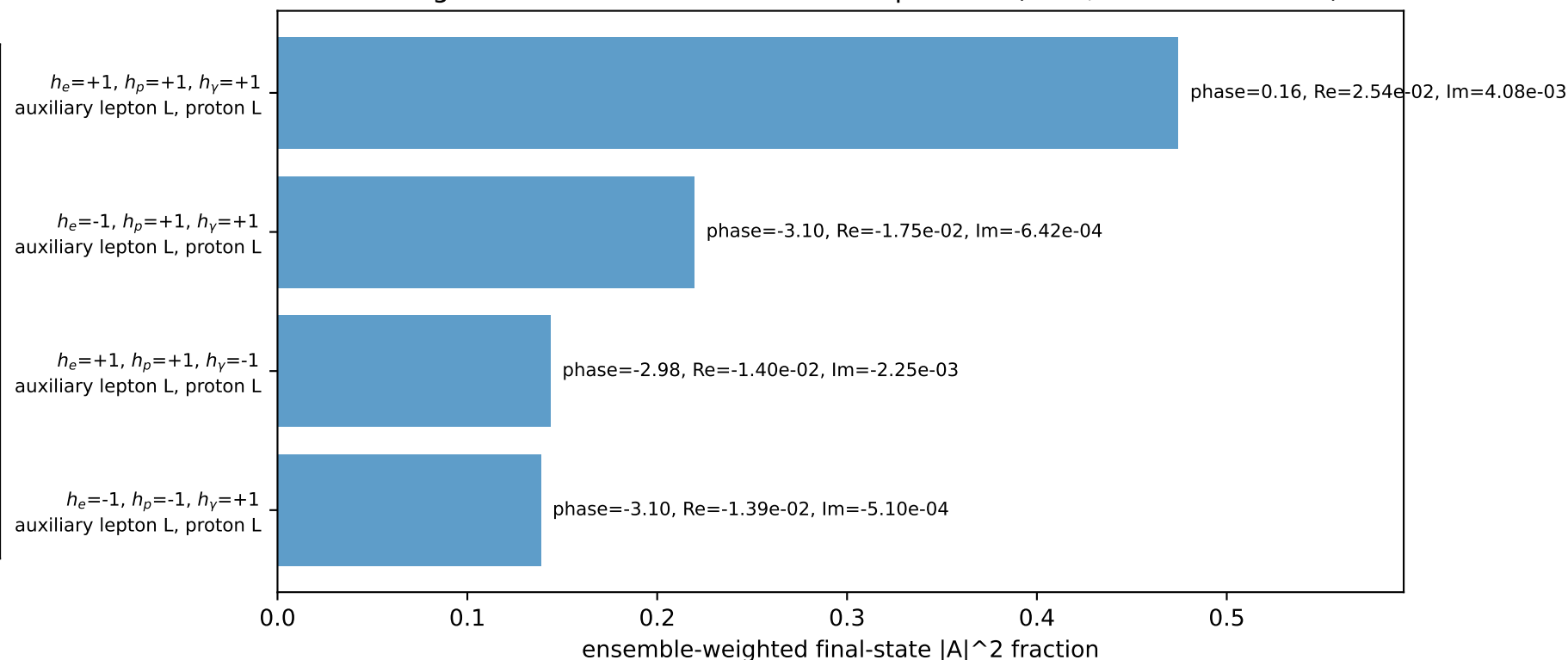
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.93673$
 $\theta_{in}=1.57079$, $\phi_l=0.0981748$, $\phi_P=3.23977$
 $E_\gamma=0.25$, $\phi_\gamma=0.19635$
 $Q^2=10.2807$, $x_B=0.920571$, $t=-8.96624$, $W^2=1.76689$, $y=0.568741$
 $\theta(l', \gamma)=108.29$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.0968$ $p_y= 0.2065$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.0968$ $p_y= -0.2065$ $p_z= 0.0000$
 l' $E= 2.2366$ $p_x= -0.2452$ $p_y= -1.9855$ $p_z= 0.0000$
 P' $E= 2.1519$ $p_x= 0.0000$ $p_y= 1.9367$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2452$ $p_y= 0.0488$ $p_z= 0.0000$

Transverse plane

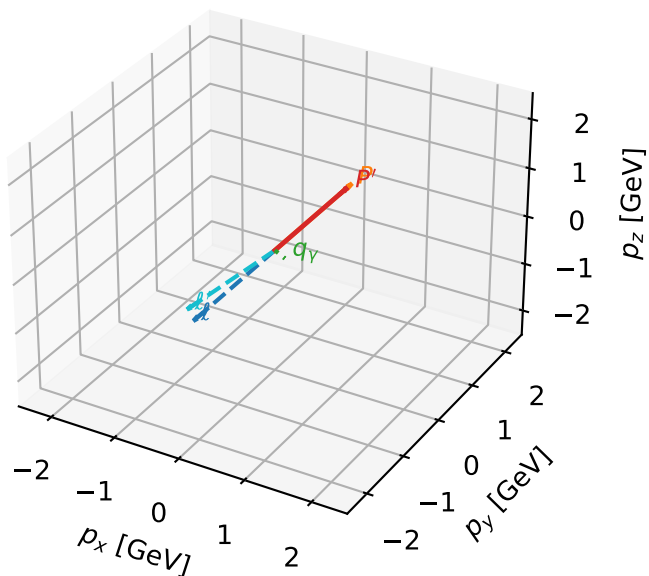


Leading initial-ensemble/final-state components (N=4, retained=97.7%)



L auxiliary lepton + L proton characteristic max F_3 configuration

3D momenta

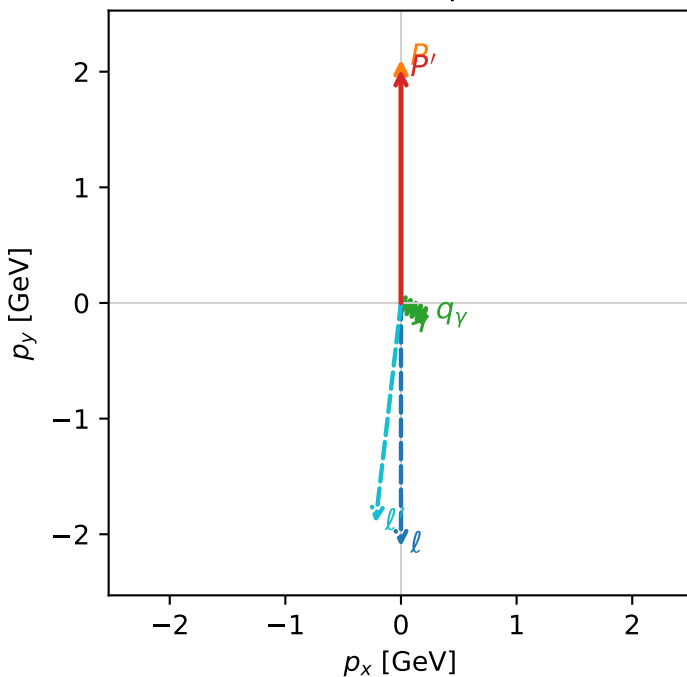


f3_low_egamma_ll_region_2 F_3=0.191658
 region: low_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\ell\rangle \otimes |\bar{L}_p\rangle$

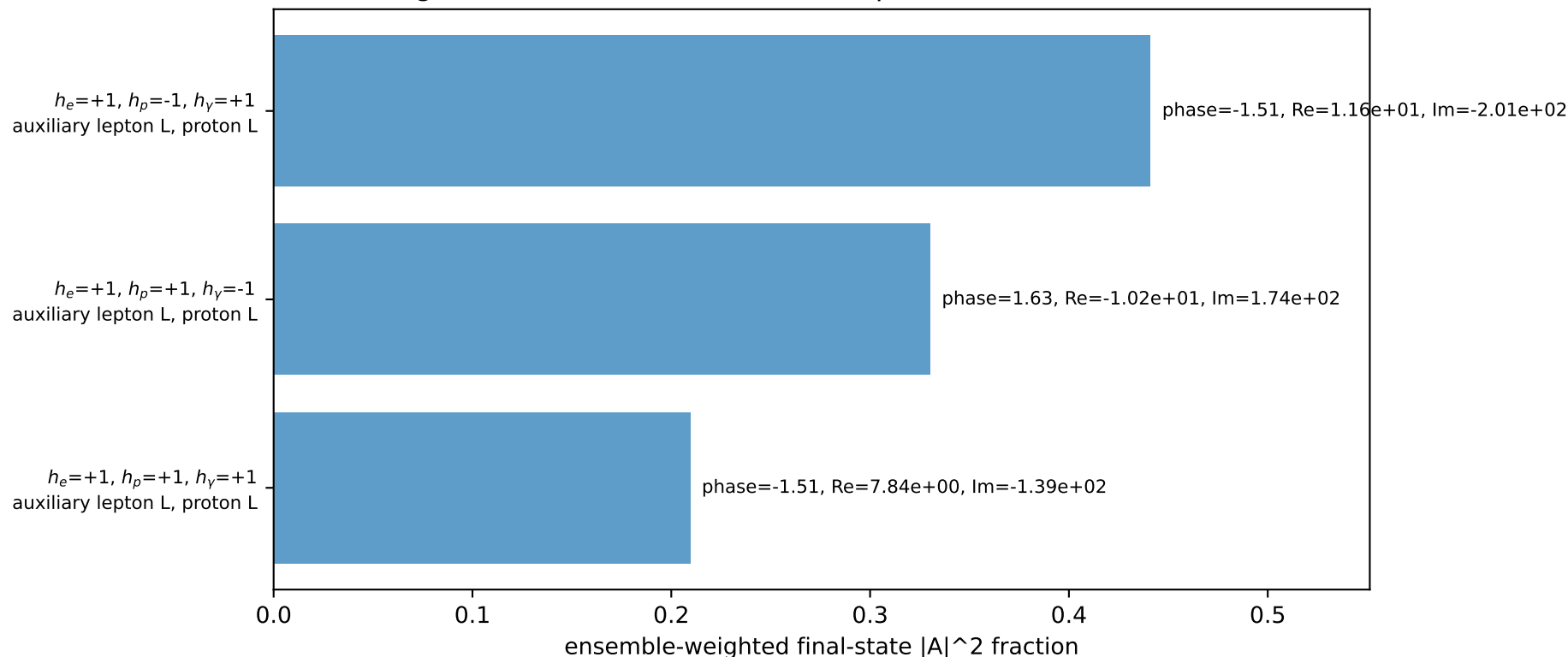
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.02061$
 $\theta_{in}=1.57079$, $\phi_\ell=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=5.79231$
 $Q^2=0.0609249$, $x_B=0.0369011$, $t=-0.0012766$, $W^2=2.46995$, $y=0.0840819$
 $\theta(\ell', \gamma)=68.4846$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.1608$ $p_x= -0.2205$ $p_y= -1.9028$ $p_z= 0.0000$
 P' $E= 2.2277$ $p_x= 0.0000$ $p_y= 2.0206$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2205$ $p_y= -0.1178$ $p_z= 0.0000$

Transverse plane

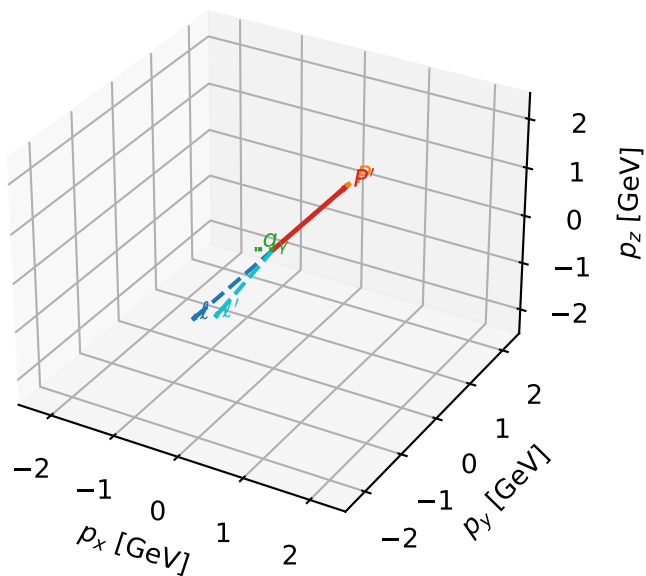


Leading initial-ensemble/final-state components (N=3, retained=98.1%)



L auxiliary lepton + L proton characteristic max F_3 configuration

3D momenta

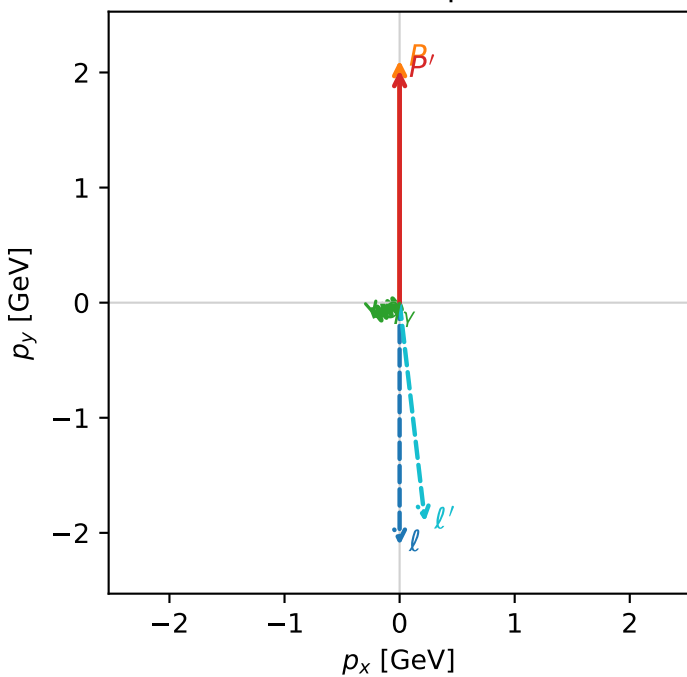


f3_low_egamma_ll_region_3 $F_3=0.191658$
 region: low_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_l\rangle \otimes |\bar{L}_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.02061$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=3.63247$
 $Q^2=0.0609249$, $x_B=0.0369011$, $t=-0.0012766$, $W^2=2.46995$, $y=0.0840819$
 $\theta(l', \gamma)=68.4846$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 l' $E= 2.1608$ $p_x= 0.2205$ $p_y= -1.9028$ $p_z= 0.0000$
 P' $E= 2.2277$ $p_x= 0.0000$ $p_y= 2.0206$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2205$ $p_y= -0.1178$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$
 auxiliary lepton L, proton L

$h_e=+1, h_p=+1, h_\gamma=-1$
 auxiliary lepton L, proton L

$h_e=+1, h_p=+1, h_\gamma=+1$
 auxiliary lepton L, proton L

Leading initial-ensemble/final-state components (N=3, retained=98.1%)

